

HW 3

To be completed by Monday, 2/23

1. Use “dataHW3 Problem1.mat” to complete this exercise. In this problem you will work with a multinomial choice model very similar to the one from the previous problem set. Upon graduating high school, individuals have four options: work ($Y=0$), attend a 2-year college ($Y=1$), attend a 4-year public college ($Y=2$), or attend a 4-year private college ($Y=3$). The available student demographic characteristics (X_i) are a constant, an indicator for whether either parent graduated from a 4-year college, and the student’s high school GPA respectively. There are two choice specific variables, distance (Z_{dist}) and price (Z_{price}) measured in thousands of dollars. Each column of these matrices reflects the choice specific attribute for the corresponding choice. In other words, the first column are the attributes for the work choice, the second column for the 2-year choice, and so on. Note that I assume that distance and price for the work option are zero.
 - (a) Estimate a multinomial logit model **assuming data on GPA is not available**. In other words, the only regressors should be the constant, parent education indicator, distance, and price. Normalize the utility of work to zero. Report parameter estimates and SEs.
 - (b) Estimate a nested logit model, **again assuming data on GPA is not available**. Put the 4-year public choice ($Y=2$) and the 4-year private choice ($Y=3$) in a nest. The work and 2-year college option are not nested. Restrict the value of λ , the nesting parameter for the 4-yr choices, to be between 0 and 1. Report parameter estimates and SEs.
 - (c) Use the delta method to calculate the standard error for λ . Does the multinomial logit model seem appropriate given the estimate of λ ?
 - (d) Estimate a multinomial logit model **including the GPA data**. The regressors will now include a constant, parent education indicator, GPA, distance, and price. Normalize the utility of work to zero. Report parameter estimates and SEs.
 - (e) Estimate a nested logit model, **again assuming data on GPA is available**. Put the 4-year public choice ($Y=2$) and the 4-year private choice ($Y=3$) in a nest. The work and 2-year college option are not nested. Restrict the value of λ to be between 0 and 1. Report parameter estimates and SEs.
 - (f) Given the results in parts (d) and (e), what was driving the correlation in the unobserved preferences for 4-yr attendance in part (b)?

2. Use “dataHW3 Problem2.mat” to complete this exercise. In this problem you will work with a simplified multinomial choice model relative to the one from the previous question. Upon graduating high school, individuals have three options: work ($Y=0$), attend a 2-year college ($Y=1$), attend a 4-year college ($Y=2$). The available student demographic characteristics (X_i) are a constant, an indicator for whether either parent graduated from a 4-year college, and the student’s high school GPA respectively. There is one choice specific variable, distance (Z_{dist}). Each column of Z_{dist} reflects the distance for the corresponding choice. In other words, the first column is the distance for the work choice, the second column for the 2-year choice, and so on. Note that I assume that distance for the work option is zero.
 - (a) Estimate a multinomial probit model. Normalize the observed and unobserved components of the work choice to zero, and assume that the variance matrix of the unobserved preferences for the 2-yr and 4-yr option is given by: $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$. Note that this is really the variance matrix of the error differences relative to the work choice. When estimating the model, set $R=300$ and $\lambda = 0.1$ (smoothness parameter in the AR Simulator).
 - The true underlying parameters are:
 - $\beta_X = [[-0.75; 0.25; 0.45] [-2.0; 0.75; 1.25]]$, the first column is for 2-yr and second column is for 4-yr
 - $\beta_Z = -0.05$
 - $\rho = 0.25$